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Studierichting: Informatica
Oefeningenreeks 4A nr. 39.4

$$I = \int \frac{3x+1}{(x^2-4x+5)^2} dx = \frac{3}{2} \int \frac{d(x^2-4x+5)}{(x^2-4x+5)^2} + 7 \int \frac{dx}{(x^2-4x+5)^2}$$

$$I = \frac{-3}{2(x^2-4x+5)} + 7 \int \frac{dx}{((x-2)^2+1)^2}$$

$$M_1 = \text{Bgtan}(x-2)$$

$$M_2 = \frac{1}{2} \text{Bgtan}(x-2) + \frac{x-2}{2(x^2-4x+5)}$$

$$I = \frac{-3}{2(x^2-4x+5)} + \frac{7}{2} \text{Bgtan}(x-2) + \frac{7(x-2)}{2(x^2-4x+5)} + C$$

$$I = \frac{7x-17}{2(x^2-4x+5)} + \frac{7}{2} \text{Bgtan}(x-2) + C$$

References